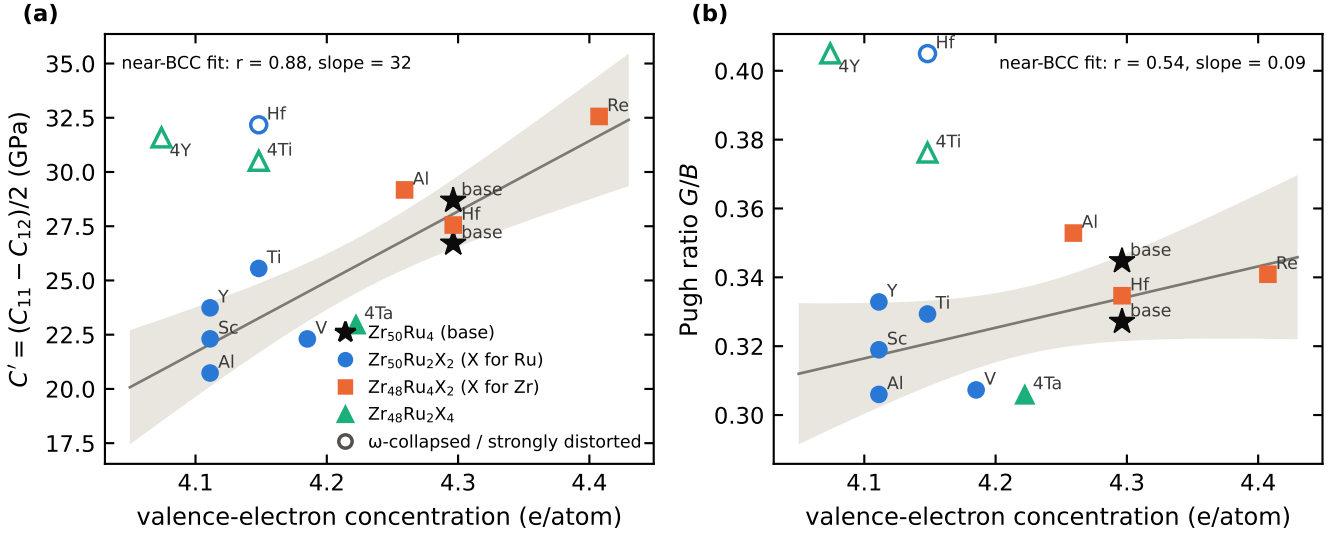
(a)
near-BCC fit: r = 0.88, slope = 32
$C' = (C_{11} - C_{12})/2$ (GPa)
valence-electron concentration (e/atom)
Hf
4Y
4Ti
Re
Al
base
Hf
base
Ti
Y
Sc
V
Al
4Ta
$Zr_{50}Ru_4$ (base)
$Zr_{50}Ru_2X_2$ (X for Ru)
$Zr_{48}Ru_4X_2$ (X for Zr)
$Zr_{48}Ru_2X_4$
ω-collapsed / strongly distorted
(b)
near-BCC fit: r = 0.54, slope = 0.09
Pugh ratio G/B
valence-electron concentration (e/atom)
Hf
4Y
4Ti
Al
base
Re
Hf
base
Y
Ti
Sc
Al
V
4Ta